

Why Rural vs Urban Was Hard to Identify

The dataset has no Rural/Urban label, so it had to be derived from housing density (sales per km²) computed directly from each zip code's lat/long coordinates.

What Didn't Work

- ✗ Lot size alone
- ✗ Distance from downtown alone
- ✗ Sales count per zip alone

No single feature gave a reliable Rural/Urban threshold.

What We Used: Housing Density

Density = sales per km² per zip, from the bounding box of its lat/long

Split found automatically at 1.04 houses/km² (largest log-scale gap)

7 of 70 zips classified Rural: 98010, 98014, 98019, 98022, 98024, 98045, 98070

✓ vs RUCA: 4/5 Rural zips matched (80% recall) • 94.3% accuracy
(vs 92.9% baseline)